We depict the graphs of both $y = \sin x,\ -\frac{\pi}{2} \le x \le \frac{\pi}{2}$ and $y = \sin^{-1} x,\ -1 \le x \le 1$ together for a better understanding.

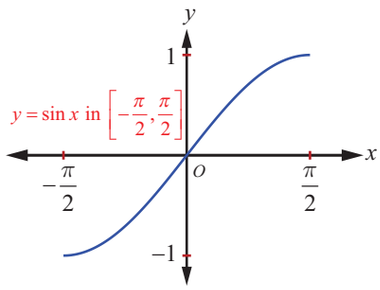


Fig. 4.7

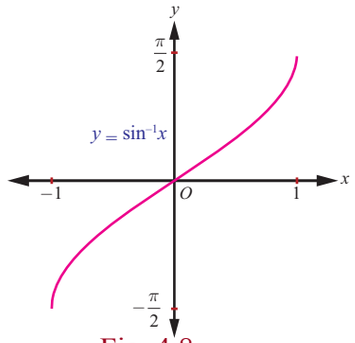


Fig. 4.8

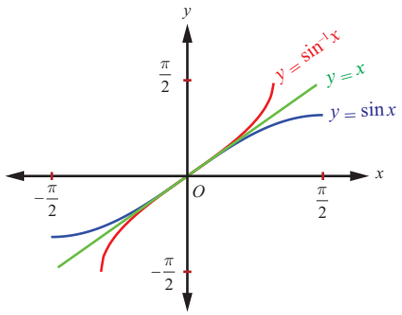


Fig. 4.9

Fig. 4.9 illustrates that the graph of $y = \sin^{-1} x$ is the mirror image of the graph of $y = \sin x,\ -\frac{\pi}{2} \le x \le \frac{\pi}{2}$, in the line $y = x$ and also shows that the sine function and the inverse sine function are symmetric with respect to the origin.

Example 4.1

Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ (in radians and degrees).

Solution

Let $\sin^{-1}\left(-\frac{1}{2}\right) = y$. Then $\sin y = -\frac{1}{2}$.

The **range** of the principal value of $\sin^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and hence, let us find $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin y = -\frac{1}{2}$. Clearly, $y = -\frac{\pi}{6}$.

Thus, the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is $-\frac{\pi}{6}$. This corresponds to $-30°$. ■

Example 4.2

Find the principal value of $\sin^{-1}(2)$, if it exists.

Solution

Since the domain of $y = \sin^{-1} x$ is $[-1,\ 1]$ and $2 \notin [-1, 1]$, $\sin^{-1}(2)$ does not exist. ■

Example 4.3

Find the principal value of

(i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (ii) $\sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right)$ (iii) $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$.

Solution

We know that $\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is given by

$\sin^{-1} x = y$ if and only if $x = \sin y$ for $-1 \le x \le 1$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$. Thus,

Chapter 4 Inverse Trigonometry.indd 136 31-01-2020 17:41:40